

Quiz (Habanero)

Q1. Simplify $\sqrt{64}$

- (A) 8
- (B) 16
- (C) 32
- (D) 4
- (E) 2

Q2. A storm brings $\sqrt{256}$ inches of rainfall. How many inches is this?

- (A) 12
- (B) 16
- (C) 20
- (D) 24
- (E) 28

Q3. What is $\sqrt[3]{125}$?

- (A) 5
- (B) 10
- (C) 15
- (D) 20
- (E) 25

Q4. A glacier melts at a rate of $\sqrt{9}$ meters per hour. How many meters melts in 2 hours?

- (A) 4
- (B) 6
- (C) 8
- (D) 12
- (E) 18

Q5. Which is equal to $\sqrt{100}$?

- (A) 2^2
- (B) 5^2
- (C) 10^2
- (D) 2^3
- (E) 5^3

Q6. A temperature drop of $\sqrt[3]{27}$ degrees Celsius occurs. How many degrees is this?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q7. What is $\sqrt{121}$?

- (A) 10
- (B) 11
- (C) 12
- (D) 13
- (E) 14

Q8. A river's flow rate is $\sqrt{49}$ cubic meters per second. How many cubic meters per second is this?

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Open Ended Questions (FRQ)

Q9. An ecosystem's population grows at a rate of $\sqrt[3]{t}$ where t is time in months. Find the growth rate after 8 months.

Q10. A rainforest's temperature variation can be modeled by the function $f(x) = \sqrt{x+4}$. Find the value of x when the temperature is $\sqrt{25}$ degrees Celsius.

Chef's Tips

- Provide a calculator for questions 9-10
- Ensure students show all work for free response questions
- Allow 30 minutes to complete the quiz